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Unit :- 3

Complex functions

Reference: "B. S. Grewal, 42nd edition"

↳ "Believing in yourself
is always a great
idea."

↳ "Great things never
come from comfort
zones"

Unit ③

Complex functions

(1)

Defn. If for each value of the complex variable $z (= x+iy)$ in a given region R , we have one or more value of $f(z)$ ($= u+iv$), then $f(z)$ is said to be a complex function & we write,

$$f(z) = u(x,y) + i v(x,y)$$

(u & v are real functions of x and y)
Here we denote $f(z)$ by w & write
 $w = f(z)$, so, we can say that

$$w = u(x,y) + i v(x,y)$$

→ Complex function of a complex variable z .

If for each value of z , $\exists$ unique value of $f(z)$ then $f(z)$ is single valued otherwise multivalued.

In this course, we are only going to discuss single valued functions.

Eg ① $f(z) = z^2 = (x^2 - y^2) + i(2xy)$
 $\Rightarrow u = x^2 - y^2$ (real part of $f(z)$)

& $v = 2xy$ (imaginary part of $f(z)$)

{ u & v are funcs of x & y }.

$$\begin{aligned}
 \textcircled{2} \quad f(z) = e^z &= e^{x+iy} = e^x (e^{iy}) \\
 &= e^x (\cos y + i \sin y) \\
 &= e^x \cos y + i e^x \sin y
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow u &= e^x \cos y \\
 v &= e^x \sin y.
 \end{aligned}$$

Above examples are the examples of complex functions. $f(z) = u + iv$ (or $w = u + iv$).

Limit of complex functions.

Defⁿ Let $f(z) = u(x, y) + i v(x, y)$, $z = x + iy$

$$z_0 = x_0 + i y_0 \quad \& \quad l = u_0 + i v_0.$$

then $\lim_{z \rightarrow z_0} f(z) = l$ if and only if

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} u(x, y) = u_0 \quad \text{and}$$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} v(x, y) = v_0.$$

Eg. If $f(z) = (x^2 + y^2) + i 2xy$

then Find $\lim_{z \rightarrow 2-i} f(z)$.

Solⁿ
we have $z_0 = 2 - i = x_0 + i y_0$
 $\Rightarrow x_0 = 2, y_0 = -1$

So $\lim_{z \rightarrow z_0} f(z) = l = u_0 + i v_0$

where $u_0 = \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} u(x, y) = \lim_{\substack{x \rightarrow 2 \\ y \rightarrow -1}} (x^2 + y^2) = 5.$

and $v_0 = \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} v(x, y) = \lim_{\substack{x \rightarrow 2 \\ y \rightarrow -1}} 2xy = -4$

$\therefore \lim_{z \rightarrow z_0} f(z) = d = u_0 + i v_0 = 5 - i4. \underline{\text{Ans}}$

Results $\lim_{z \rightarrow \infty} f(z) = \lim_{z \rightarrow 0} f(1/z) \quad \text{--- (1)}$

Eg $\lim_{z \rightarrow \infty} \frac{1}{z^2} = \lim_{z \rightarrow 0} z^2 = 0.$

Eg $\lim_{z \rightarrow 2i} (3x + iy^2) = d = u_0 + i v_0$ where
 $u_0 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 2}} 3x, v_0 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 2}} y^2$

$\therefore d = 4i \quad \underline{\text{Ans}}$

Eg $\lim_{z \rightarrow \infty} \sqrt{z-2i} - \sqrt{z-i}$
 $= \lim_{z \rightarrow \infty} \frac{(\sqrt{z-2i} - \sqrt{z-i})(\sqrt{z-2i} + \sqrt{z-i})}{(\sqrt{z-2i} + \sqrt{z-i})} \xrightarrow{\text{By (1)}} -i$
 $= \lim_{z \rightarrow \infty} \frac{-i}{\sqrt{z-2i} + \sqrt{z-i}} = \lim_{z \rightarrow 0} \frac{-i}{\sqrt{\frac{1}{z}-2i} + \sqrt{\frac{1}{z}-i}}$

$$= \lim_{z \rightarrow 0} \frac{-i\sqrt{z}}{\sqrt{1-2iz} + \sqrt{1-iz}} = \frac{0}{2} = 0.$$

Continuity of complex functions

A function $f(z)$ is said to be continuous at $z \rightarrow z_0$ if following three conditions are satisfied.

(i) $f(z_0)$ exists.

(ii) $\lim_{z \rightarrow z_0} f(z)$ exists.

(iii) $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Eg. Discuss the continuity of the following funcⁿ $f(z)$ at $z=0$.

$$f(z) = \frac{x^3 - y^3}{x^2 + y^2} + i \frac{x^3 + y^3}{x^2 + y^2}, \quad z \neq 0$$

$$0, \quad z = 0.$$

Solⁿ. Here $f(z) = u + iv$

where $u = \frac{x^3 - y^3}{x^2 + y^2}$, $v = \frac{x^3 + y^3}{x^2 + y^2}$

$$\lim_{\substack{z \rightarrow 0 \\ y \rightarrow 0}} u = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3 - y^3}{x^2 + y^2}, \quad \text{let } y = mx$$

$$= \lim_{x \rightarrow 0} \frac{x^3 - m^3 x^3}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{x(1 - m^3)}{1 + m^2} = 0$$

Similarly $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} v = 0$

$$\Rightarrow u_0 = 0 \text{ \& } v_0 = 0$$

$$\Rightarrow l = 0$$

$$\Rightarrow \lim_{z \rightarrow z_0} f(z) = 0 = f(z_0).$$

$\Rightarrow f$ is continuous at $z = z_0 = 0$.

Differentiability of complex functions

A complex function $f(z)$ is said to be differentiable if $\lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}$ is

same throughout, "no matter how δz approaches to 0". Then this limit is equal to $f'(z)$.

Eg $f(z) = \bar{z} = x - iy$

$$\delta x = \delta(x + iy) = \delta x + i \delta y$$

$$z + \delta z = (x + \delta x) + i(y + \delta y)$$

$$\frac{f(x + \delta x) - f(x)}{\delta z} = \frac{\overline{(x + \delta x)} - \bar{x}}{\delta z}$$

$$= \frac{(x + \delta x) - i(y + \delta y) - (x - iy)}{\delta z}$$

$$= \frac{\delta x - i \delta y}{\delta x + i \delta y}$$

Through real axis i.e., if $\delta y = 0$ then
 $\delta z = \delta x$.

$$\lim_{\delta z \rightarrow 0} \frac{f(x+\delta z) - f(z)}{\delta z} = \lim_{\delta x \rightarrow 0} \frac{\delta x - i\delta y}{\delta x + i\delta y} = -1 \quad (1)$$

Through imaginary axis i.e., if $\delta x = 0$, then

$$\delta x = \delta y$$

$$\lim_{\delta z \rightarrow 0} \frac{f(z+\delta z) - f(z)}{\delta z} = \lim_{\delta y \rightarrow 0} \frac{\delta x - i\delta y}{\delta x + i\delta y} = 1 \quad (2)$$

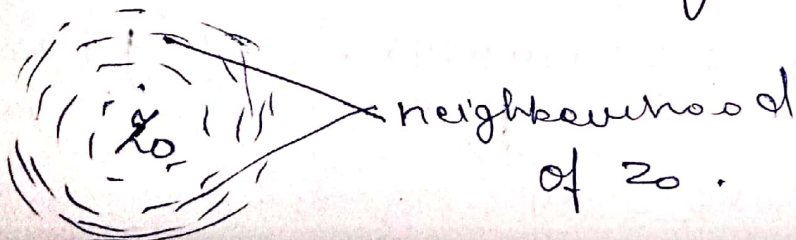
By (1) & (2), limits are different.
 $\therefore f(z)$ is not differentiable.

From now, we begin the most important part of unit (3), which is also very important from exam point of view.

Analytic functions

If $w = f(z)$ has derivative at $z = z_0$ and at all points in the neighbourhood of z_0 , then $f(z)$ is analytic at z_0 .

(Analytic or holomorphic or regular all are same thing)



Cauchy's Riemann equations (used to check analyticity of f^n).

Let $f(z) = u + iv$ then $f(z)$ is said to be analytic in D

if (i) u_x, u_y, v_x, v_y are continuous on every point of D .

(ii)
$$\left. \begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned} \right\} \text{Cauchy Riemann equations OR C-R equations}$$

and then
$$f'(z) = u_x + i v_x$$

Eg $f(z) = z\bar{z} = |z|^2$. check whether analytic or not?

Here $f(z) = |z|^2 = x^2 + y^2$

$$\Rightarrow u = x^2 + y^2$$

$$v = 0$$

$$u_x = 2x, \quad v_x = 0 = v_y$$

$$u_y = 2y$$

$$u_x = v_y \Rightarrow 2x = 0 \Rightarrow x = 0$$

$$u_y = -v_x \Rightarrow 2y = 0 \Rightarrow y = 0$$

Since u_x, u_y are linear polynomials so continuous. v_x, v_y being constants are continuous.

And C-R equations are satisfied at $x=0, y=0$.

only.

$\therefore f(z) = |z|^2$ is analytic at $(0,0)$ only.
(or at $z=0$)

Eg. If $w = \log z$, find $\frac{dw}{dz}$ and
determine whether w is analytic or not.

Soln we have, $w = u + iv$
 $= \log z$
 $= \log(x + iy)$
 $= \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} y/x$

$$\Rightarrow u = \frac{1}{2} \log(x^2 + y^2)$$

$$v = \tan^{-1} y/x.$$

(Using $\log z$
 $= \log|z| + i \arg z.$)

$$u_x = \frac{x}{x^2 + y^2}, \quad u_y = \frac{y}{x^2 + y^2}$$

$$v_x = \frac{-y}{x^2 + y^2}, \quad v_y = \frac{x}{x^2 + y^2}$$

Property ① If $f(z)$ & $g(z)$ are continuous
in D then $\frac{f(z)}{g(z)}$ is continuous everywhere
in D except at the point where $g(z)$ is 0

⑤ If $f(z)$ & $g(z)$ are continuous then $f(z) \pm g(z)$, $f(z) \cdot g(z)$ are continuous. ⑤

⑥ Polynomial functions, exponential f^n s, trigonometric func^s are everywhere continuous. $\therefore$ C.R. equations are satisfied

By property ①, u_x, u_y, v_x, v_y are continuous everywhere except at $(0,0)$

$\therefore w = \log z$ is analytic everywhere except at $(0,0)$ (or $z=0$).

$$\begin{aligned} \& \frac{dw}{dz} = f'(z) &= u_x + i v_x \\ &= \frac{x}{x^2+y^2} + i \frac{(-y)}{x^2+y^2} \\ &= \frac{x-iy}{x^2+y^2} = \frac{\bar{z}}{|z|^2} \end{aligned}$$

$$= \frac{\bar{z}}{z\bar{z}} = 1/z, \quad z \neq 0 \quad \underline{\text{Ans.}}$$

Ex-1. Find c_1 & c_2 if function $f(z) = x^2 + c_1 y^2 - 2xy + i(c_2 x^2 - y^2 + 2xy)$ is analytic.

Ex-2. Prove $\sinh z$ is analytic & find its derivative.

C.R. equations in polar form

$$z = x + iy$$

$$\text{let } x = r \cos \theta, \quad y = r \sin \theta$$

$$z = re^{i\theta}$$

$$w = f(z) = f(re^{i\theta}) = u + iv \\ = u(r, \theta) + i v(r, \theta)$$

$$\text{C.R. equations are } u_r = \frac{1}{r} v_\theta, \quad v_r = -\frac{1}{r} u_\theta$$

$$\& \quad \boxed{f'(z) = e^{-i\theta} \frac{df}{dz}} \quad \text{--- } \textcircled{*}$$

$$\text{eg } f(z) = f(re^{i\theta}) = r \cos \theta + i r \sin \theta$$

$$\Rightarrow u = r \cos \theta \\ v = r \sin \theta$$

$$u_r = \cos \theta, \quad v_r = \sin \theta, \quad u_\theta = -r \sin \theta, \quad v_\theta = r \cos \theta$$

for $f(z)$ All are trigonometric functions $\therefore$

Continuous

& C.R. equations are satisfied

$\therefore f(z)$ is analytic.

$$f'(z) = \cancel{u_r + i v_r} = e^{-i\theta} (\cos \theta + i \sin \theta) \\ = e^{-i\theta} (e^{i\theta}) \\ = 1$$

$$\uparrow \text{ Using } \textcircled{*} \& \cdot e^{i\theta} = \cos \theta + i \sin \theta$$

Ex find $u(x, \theta)$ if $f(z) = u(x, \theta) + i v(x, \theta)$
 is analytic such that $v(x, \theta) = x^2 \cos 2\theta - x \cos \theta + 2$.

Soln f is analytic implies $u_x = \frac{1}{x} v_\theta$ — (1)

$$v_x = -\frac{1}{x} u_\theta \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow u_x = \frac{1}{x} (v_\theta) = -2x \sin 2\theta + \sin \theta.$$

$$\Rightarrow \frac{du}{dx} = -2x \sin 2\theta + \sin \theta.$$

$$\Rightarrow du = (-2x \sin 2\theta + \sin \theta) dx$$

$$\Rightarrow \int du = \int (-2x \sin 2\theta + \sin \theta) dx$$

$$\Rightarrow u = -x^2 \sin 2\theta + x \sin \theta + c(\theta) \quad \text{--- (3)}$$

(since we integrated w.r. to x
 treating θ as constant
 so constant fn c
 will be a fn of θ i.e., $c(\theta)$)

To find $c(\theta)$

$$\text{By (3)} \quad u_\theta = -x^2 \cos 2\theta + x \cos \theta + c'(\theta) \quad \text{--- (4)}$$

$$\text{Also by (2)} \quad u_\theta = -x v_x \\ = -2x^2 \cos 2\theta + x \cos \theta \quad \text{--- (5)}$$

equating (4) & (5), we get $c'(\theta) = 0 \Rightarrow c(\theta) = K$
 Constant

$$\Rightarrow u = -x^2 \sin 2\theta + x \sin \theta + k$$

$$\Rightarrow f = u + iv$$

$$= (-x^2 \sin 2\theta + x \sin \theta + k) + i(x^2 \cos 2\theta - x \cos \theta + 2)$$

Ans.

Properties of analytic functions

① If f is analytic then $f'(z) = u_x + i v_x$

② If $f = u + iv$ is analytic then u and v are harmonic.

③ If $f = u + iv$ is analytic then $u = C_1$, $v = C_2$ are orthogonal families of curves.

Eg If $f(z)$ is analytic f^n of z show that

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] [u(x,y)]^2 = 2 |f'(z)|^2 \quad \text{--- (1)}$$

Soln f is analytic $\Rightarrow f'(z) = u_x + i v_x$ &
 $u_x = v_y, u_y = -v_x$

$$\therefore |f'(z)|^2 = u_x^2 + v_x^2$$

$$\text{L.H.S. of (1)} = \frac{\partial^2}{\partial x^2} u^2 + \frac{\partial^2}{\partial y^2} u^2$$

$$\begin{aligned} \text{now, } \frac{\partial^2}{\partial x^2} (u^2) &= \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} u^2 \right] = \frac{\partial}{\partial x} \left[2u \cdot \frac{\partial u}{\partial x} \right] \\ &= 2u \left(\frac{\partial^2 u}{\partial x^2} \right) + 2 \frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial x} \end{aligned}$$

Similarly $\frac{\partial^2 (v^2)}{\partial y^2} = 2v \frac{\partial^2 v}{\partial y^2} + 2 \left(\frac{\partial v}{\partial y} \right)^2$ (2)

$$\begin{aligned} \therefore \text{L.H.S of (1)} &= 2u \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right] + 2[u_x^2 + u_y^2] \\ &= 2[u_x^2 + u_y^2] \\ &= 2[u_x^2 + (-v_x)^2] \\ &= 2(u_x^2 + v_x^2) = 2|f'(z)|^2 \\ &= \text{R.H.S of (1).} \end{aligned}$$

Milne Thompson method (or M-T method)

(Used to write $f(z)$ when either of u or v is given).

Replace x by z & y by 0 in $f'(z)$ to get $f'(z)$ in terms of z then integrate it to obtain $f(z)$.

Eg ① Determine an analytic function $f(z)$ whose real part is $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 2x + 1$

Also prove that it satisfies Laplace eqn.

Defⁿ $\Rightarrow$ Harmonic functions :- A function $f(x, y)$ in which $f_x, f_y, f_{xx}, f_{yy}, f_{xy}$ are continuous and satisfy Laplace eqn $\rightarrow f_{xx} + f_{yy} = 0$, is called Harmonic. f' .

eg. $f(x,y) = x^2y^2 - x - 2$

$f_x = 2x$, $f_y = -2y$, $f_{xx} = 2$,

$f_{yy} = -2$, $f_{xy} = 0$ are all continuous being constant & linear polynomials.

Also $f_{xx} + f_{yy} = 2 - 2 = 0 \therefore f$ is a Harmonic function. $\therefore$

Solⁿ of Eg. ① Given that

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 2x + 1$$

Since f is analytic $\Rightarrow f'(z) = u_x + i v_x$

and $u_x = v_y$, $u_y = -v_x$.

$$u_x = 3x^2 - 3y^2 + 6x + 2$$

$$v_x = -u_y = 6xy + 6y$$

$$\therefore f'(z) = u_x + i v_x = (3x^2 - 3y^2 + 6x + 2) + i(6xy + 6y) \quad \text{--- ①}$$

Replacing x by z & y by 0 } Using M-T method
in ①, we get

$$f'(z) = (3x^2 + 6x + 2) + 0i$$

Integrating $\Rightarrow \boxed{f(z) = z^3 + 3x^2 + 2x + C}$

7. $v = \log(x^2 + y^2) + x - 2y$, find analytic funcⁿ $f(z)$. (8)

Solⁿ $f(z) = u + i v$

$$v_x = \frac{2x}{x^2 + y^2} + 1$$

$$u_x = v_y = \frac{2y}{x^2 + y^2} - 2$$

$$f(z) = \frac{2y}{x^2 + y^2} - 2 + i \left(\frac{2x}{x^2 + y^2} + 1 \right) \quad \text{--- (1)}$$

Replacing x by z & y by 0 in (1)

$$f(z) = i \left(\frac{2}{z} + 1 \right) - 2 \quad \text{--- (2)}$$

$$\Rightarrow f(z) = 2i \log z + iz - 2z + C \quad \text{(By integrating (2))}$$

Eg If $u + v = \frac{2 \sin 2x}{e^{2y} + e^{-2y} - 2 \cos 2x}$ --- (1) and

$f(z) = u + i v$ is analytic fⁿ of z find $f(z)$.

Solⁿ Differentiating (1) partially w.r.to x and y respectively

As $u + v = \frac{\sin 2x}{\cosh 2y - \cos 2x} \quad \left[\frac{e^{2y} + e^{-2y}}{2} = \cosh 2y \right]$

$$u_x + v_x = \frac{2 \cosh 2y \cos 2x - 2}{(\cosh 2y - \cos 2x)^2} \quad - (2)$$

$$u_y + v_y = - \frac{2 \sinh 2y \sin 2x}{(\cosh 2y - \cos 2x)^2} \quad - (4)$$

$$\Rightarrow u_x - v_x = - \frac{2 \sinh 2y \sin 2x}{(\cosh 2y - \cos 2x)^2} \quad - (3)$$

Solving (2) & (3), (Using $u_y = -v_x$
we get $u_x = v_y$ in (4))

$$u_x = \frac{(\cosh 2y \cos 2x - \sinh 2y \sin 2x - 1)}{(\cosh 2y - \cos 2x)^2}$$

$$v_x = \frac{(\cosh 2y \cos 2x + \sinh 2y \sin 2x - 1)}{(\cosh 2y - \cos 2x)^2}$$

$$f'(z) = u_x + i v_x$$

$$= \frac{(\cosh 2y \cos 2x - \sinh 2y \sin 2x - 1) + i (\cosh 2y \cos 2x + \sinh 2y \sin 2x - 1)}{(\cosh 2y - \cos 2x)^2} \quad - (4)$$

By Milne Thompson method in (4),
we get,

$$f'(z) = \frac{(\cos 2z - 1) + i(\cos 2z - 1)}{(1 - \cos 2z)^2} \quad (9)$$

$$= \frac{(1+i)}{\cos 2z - 1} = -(1+i) \cdot 2 \sin^2 z$$

$$= \frac{-(1+i)}{2} \operatorname{cosec}^2 z$$

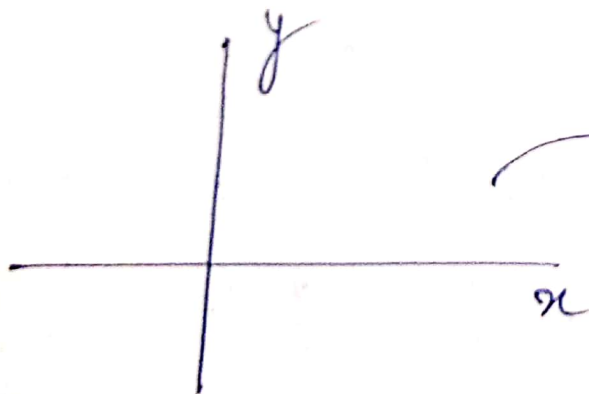
$$f(z) = \frac{1}{2} (1+i) \cot z$$

, integrating we get

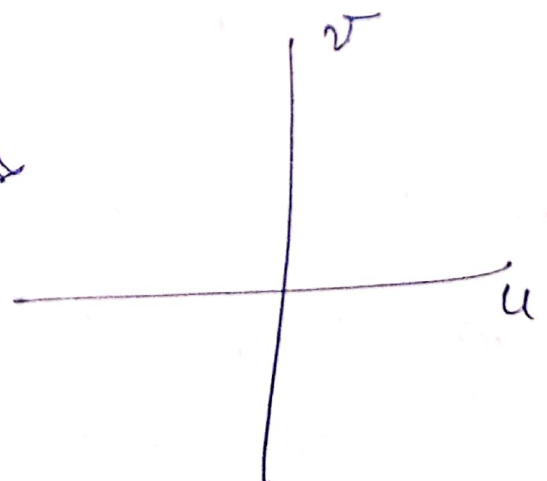
Conformal mapping

we have complex z & $w = f(z)$.

for given values of z , we get same value of w (or $f(z)$)



z -plane

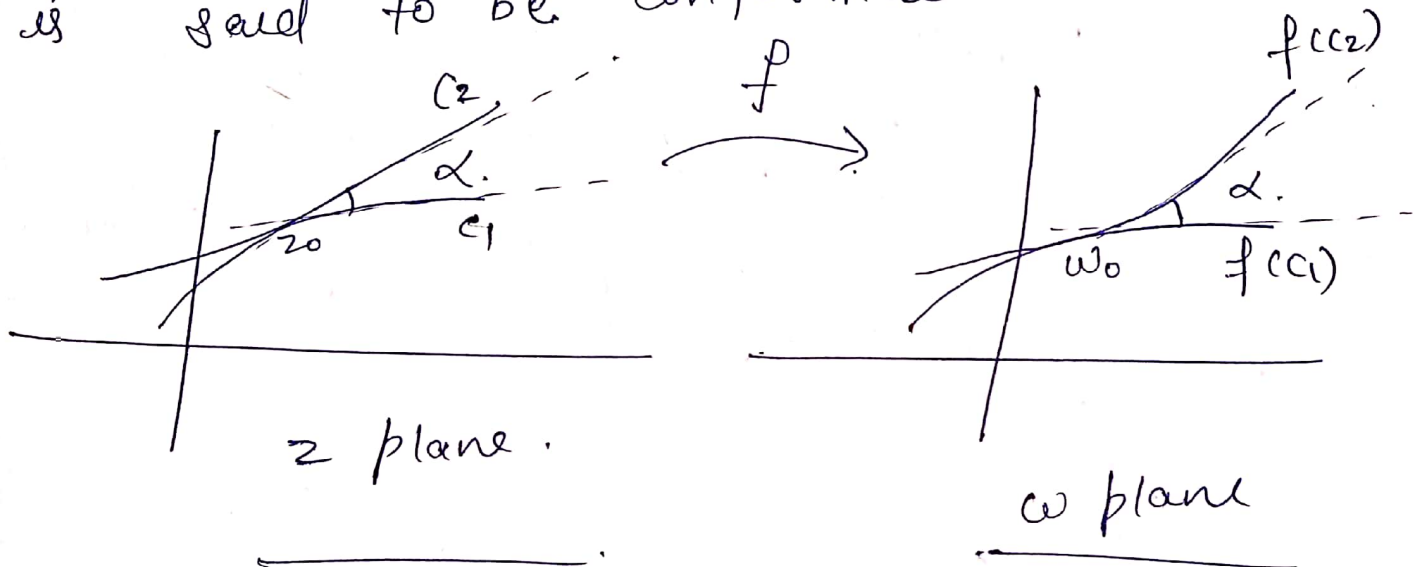


w plane.

i.e., given values of z in z plane, we obtain corresponding values of w in w plane. (or $f(z)$)

Consider two curves C_1 & C_2 in z -plane
 Suppose angle b/w them is α at their intersection point z_0 .
 Let $f(C_1)$ & $f(C_2)$ are the image curves
 of C_1 & C_2 in w -plane.

If the corresponding image curves
 $f(C_1)$ & $f(C_2)$ in w -plane intersect
 at the point w_0 at the same angle
 α and if the sense of rotation is
 also preserved then such mapping f
 is said to be conformal.



Result If $f(z)$ is analytic then $f(z)$
 is conformal where $f'(z) \neq 0$.

Eg $f(z) = z^2 + 2z$
 then being a polynomial $f(z)$
 is analytic, Also $f'(z) = 2z + 2 \neq 0$
 if $z \neq -1$
 $\therefore f$ is conformal $\forall z \neq -1$.

Let $z = re^{i\theta}$, $w = f(z) = \rho e^{i\phi}$

(10)

Then $\frac{\rho}{r} = \text{coeff. of magnification}$

$\phi - \theta = \text{angle of rotation}$

Ex for a conformal mapping $w = z^2$ show that

(i) coeff. of magnification at $z = 2+i$ is $\sqrt{5}$.

(ii) Angle of rotation at $z = 2+i$ is $\tan^{-1}(0.5)$

Soln (i) let $z = 2+i = re^{i\theta}$
 $\Rightarrow r = \sqrt{2^2+1} = \sqrt{5}$

$$\theta = \tan^{-1} 0.5.$$

~~$w = z^2 = (re^{i\theta})^2 = r^2 e^{i2\theta}$~~

$$w = z^2 \Rightarrow \rho e^{i\phi} = (re^{i\theta})^2$$

$$\Rightarrow \rho e^{i\phi} = r^2 e^{2i\theta} \quad \text{--- (1)}$$

$$\Rightarrow \rho = r^2 = 5.$$

$$\therefore \frac{\rho}{r} = \frac{5}{\sqrt{5}} = \sqrt{5}.$$

(ii) By (1) $\phi = 2\theta = 2 \tan^{-1}(0.5)$

$$\phi - \theta = 2\theta - \theta = \theta = \tan^{-1}(0.5) \quad \text{Ans.}$$

eg Find the image of the infinite strip $\frac{1}{4} \leq y \leq \frac{1}{2}$ under the transform $w = \frac{1}{z}$. Also show that region graphically.

Soln

$$w = \frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)}$$

$$= u+iv$$

$$\Rightarrow u = \frac{x}{x^2+y^2}, \quad v = \frac{-y}{x^2+y^2} \quad \text{--- (1)}$$

$$\Rightarrow \frac{y}{v} = -\frac{x}{u} \quad \text{or} \quad x = -\frac{uy}{v}$$

Let $y = a$ then $x = -\frac{au}{v}$ so,

By (1)
$$v = \frac{-a}{x^2+y^2} = \frac{a}{\frac{a^2u^2}{v^2} + a^2}$$

$$\Rightarrow u^2 + v^2 + \frac{v}{a} = 0 \quad \text{--- (2)}$$

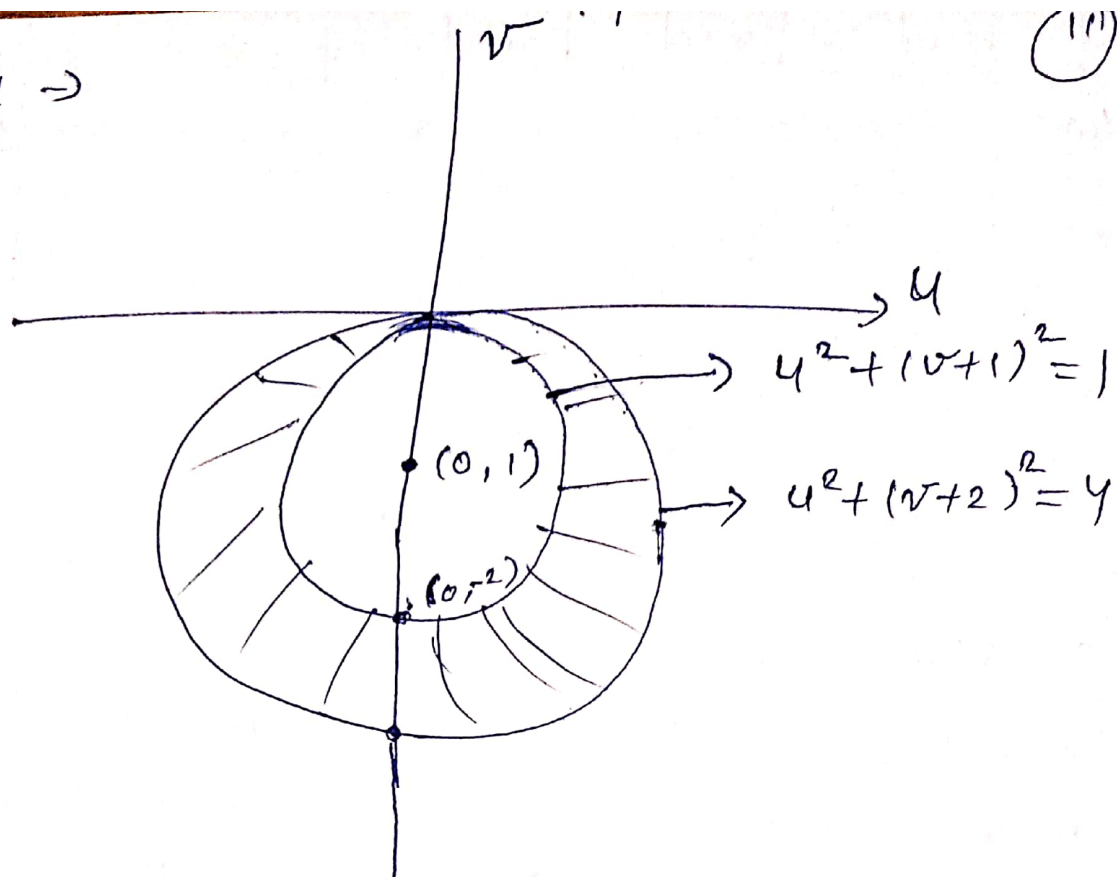
for $y = \frac{1}{2}$ (2) $\Rightarrow u^2 + v^2 + 2v = 0$

$$\Rightarrow u^2 + (v+1)^2 = 1 \quad \text{--- (3)}$$

for $y = \frac{1}{4}$ (2) $\Rightarrow u^2 + v^2 + 4v = 0$

$$\Rightarrow u^2 + (v+2)^2 = 4 \quad \text{--- (4)}$$

Graphically $\rightarrow$



Bilinear transformation

The transformⁿ $w = \frac{az+b}{cz+d}$, a, b, c, d are complex constants & $ad-bc \neq 0$ is known as bilinear transformation. clearly this map is analytic and

$f'(z) = \frac{ad-bc}{(cz+d)^2} \neq 0 \therefore$ it is conformal.

Also inverse bilinear map $z = \frac{-dw+b}{cw-a}$ is also conformal by same method.

Fixed point of Bilinear map.

A point $z = z_0$ is said to be fixed pt. or invariant pt. of Bilinear map if $f(z_0) = z_0$ i.e., if $\frac{az_0+b}{cz_0+d} = z_0$.

Eg. find fixed point of $w = \frac{z+i}{2z+1}$

let $f(z_0) = z_0$

$$\Rightarrow \frac{z_0+i}{2z_0+1} = z_0$$

$$\Rightarrow z_0+i = 2z_0^2+z_0$$

$$\Rightarrow 2z_0^2 = i \Rightarrow z_0 = \pm \sqrt{\frac{i}{2}}$$

Eg. Show that the relation $w = \frac{5-4z}{4z-2}$ transforms the circle $|z|=1$ into the circle of radius 1 in w -plane & find the centre of the circle.

Soln $w = \frac{5-4z}{4z-2}$ is a bilinear map.

$$\Rightarrow z = \frac{2w+5}{4w+4}$$

$$|z|=1 \Rightarrow \left| \frac{2w+5}{4w+4} \right| = 1$$

$$\Rightarrow |2w+5| = |4w+4|$$

$$\Rightarrow |2u+2iv+5| = |4u+4iv+4| \quad \text{as } w = u+iv$$

$$\Rightarrow (2u+5)^2 + 4v^2 = (4u+4)^2 + 16v^2$$

$$\Rightarrow \left(u + \frac{1}{2}\right)^2 + v^2 = 1$$

which is a circle in w plane with centre $(-\frac{1}{2}, 0)$ & radius 1.

Cross ratio

Cross ratio of 4 variables (z_1, z_2, z_3, z_4) is defined as $\frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)}$. — (1)

If $w_i = \frac{az_i + b}{cz_i + d}$ ($i = 1, 2, 3$).

Bilinear transfⁿ under the points z_1, z_2, z_3 in z plane are mapped to the points w_1, w_2, w_3 in w -plane as given by,

$$\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)} \quad \text{--- (2)}$$

Eg find the Bilinear transfⁿ which maps the points $i, -i, 1$ of z plane into $0, 1, \infty$ of the w -plane respectively.

Solⁿ let $z_1 = i, z_2 = -i, z_3 = 1$
∴ $w_1 = 0, w_2 = 1, w_3 = \infty$

then By (2) $\frac{(w - 0)(1 - w_3)}{(w - w_3)(1 - 0)} = \frac{(z - i)(-i - 1)}{(z - 1)(-i - i)}$

$$\Rightarrow \frac{w \left(\frac{1}{w_3} - 1 \right)}{\left(\frac{w}{w_3} - 1 \right)} = \frac{(z - i)(-i - 1)}{(z - 1)(-2i)}$$

$$\Rightarrow \frac{w(-1)}{(-1)} = \frac{(1+i)(z-i)}{2i(z-1)}$$

$$\left[\begin{array}{l} \because w_3 = \infty \\ \Rightarrow \frac{1}{w_3} \rightarrow 0 \end{array} \right]$$

$$\Rightarrow w = -\frac{(i+1)}{2} \frac{(z-i)}{(z-1)}$$

Cauchy's integral theorem

If $f(z)$ is analytic and its derivative $f'(z)$ is continuous at all points inside and on a closed curve C then $\oint_C f(z) dz = 0$.

Eg Evaluate $\oint_C z^2 dz$ using Cauchy's

integral theorem where $C: |z|=1$.

Solⁿ we have $f(z) = z^2 = (x+iy)^2$
 $= x^2 - y^2 + 2ixy$

$$\Rightarrow u = x^2 - y^2, v = 2xy$$

C-R. eqⁿs are clearly satisfied $\therefore$

It is analytic & $f'(z) = 2z$ is continuous

$$\Rightarrow \oint_C f(z) dz = 0 \quad (\text{by above theorem})$$

Cauchy's integral formula

If $f(z)$ is analytic within & on a closed curve C & if a is any point within C , then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)} dz \quad - (1)$$

and $f^n(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z) \cdot dz}{(z-a)^{n+1}} \quad \text{--- (2)}$

(13)

Eg. Evaluate $\oint_C \frac{z^2+1}{z(z+1)} dz$. $C: |z|=1$.

Solⁿ. Let $I = \oint_C \frac{z^2+1}{z(z+1)} dz$

$$= \oint_C (z^2+1) \left(\frac{1}{z} - \frac{2}{2z+1} \right) dz$$

$$= \oint_C \frac{z^2+1}{z} - \oint_C \frac{2(z^2+1)}{2(z+\frac{1}{2})} dz$$

$$= 2\pi i \cdot f(0) - 2\pi i \cdot f(-\frac{1}{2}) \quad (\text{By (1)})$$

$$= 2\pi i - 2\pi i \cdot \frac{5}{4}$$

$$= -\frac{\pi i}{2}$$

Eg. Evaluate $\oint_{|z|=3} \frac{2z^2 - z - 3}{(z-2)^2} dz$.

Here $f(z) = 2z^2 - z - 3$.

$$= \oint \frac{f(z)}{(z-2)^{2+1}} dz = \frac{2\pi i}{2!} f^{(2)}(2) \quad (\text{by (2)})$$

$$= 4\pi i \quad \text{Ans.}$$

Ex → ① $I = \int_C \frac{e^z dz}{(z^2 + \pi^2)^2}$ $C: |z| = 4$
Ans = i/π .

② $I = \int_C \frac{e^z dz}{z(1-z)^3}$: $C: |z-1| = 1/2$
Ans = $-\pi i e$.

Zeros of an analytic function :-

z_0 is a zero of analytic function $f(z)$
 if $f(z_0) = 0$.

if $f(z_0) = f'(z_0) = f''(z_0) = \dots = f^{(n-1)}(z_0) = 0$

but $f^{(n)}(z_0) \neq 0$

then z_0 is zero of order n of f .

eg. $f(z) = z^2 \sin z$

$f(0) = 0$

$f'(0) = 0$

$f''(0) = 0$

$f'''(0) \neq 0$

} check.

$\therefore 0$ is a zero of order 3 of f .

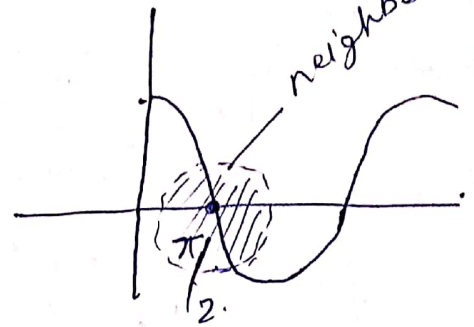
↳ Zeros of analytic functions are isolated.

i.e., if z_0 is a zero of analytic $f^n \neq 0$ then there exist a neighbourhood around

z_0 , such that in that neighbourhood (14) no zero of f will exist.

Eg $f(z) = \cos z$. $z_0 = \pi/2$.

z_0 is an isolated zero of $\cos z$.



Singularities

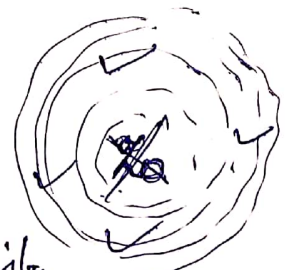
A point z_0 at which $f(z)$ is not analytic is known as singular point or singularity of $f(z)$.

Types of singularities

(a) isolated singularity :-

If $f(z)$ is analytic at every pt in the neighbourhood of a point z_0 except at z_0 itself. then z_0 becomes isolated singularity of f .

Eg a) $f(z) = 1/z$, $z=0$ is isolated singularity



b) $f(z) = \frac{1}{z^3(z^2+1)}$

$z=0, \pm i$ are isolated singularity

b) Removable singularity :-
 If $f(z)$ is not defined at $z = z_0$
 but $\lim_{z \rightarrow z_0} f(z)$ exists then $z = z_0$ is
 called removable singularity.

Eg $f(z) = \frac{\sin z}{z}$

not defined at $z = 0$.

but $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$

$\Rightarrow z = 0$ is removable singularity.
 we can redefine $f(z)$ as

$$f(z) = \begin{cases} \frac{\sin z}{z}, & z \neq 0 \\ 1, & z = 0 \end{cases}$$

Now f becomes analytic without any
 singularity.

Taylor's series expansion

If $f(z)$ is analytic inside a circle C
 with centre a then Taylor's expansion
 about a is

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(z-a)^n + \dots \quad \text{--- (1)}$$

Eg. $f(z) = \sin z$ ~~at $z=0$~~ about $z=0$ (15)

Taylor's series expansion is given
by, $\sin z = \frac{z}{1!} - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots$ (by (1))

Eg obtain the Taylor's expansion of

$$f(z) = \frac{1}{z^2 + (1+2i)z + 2i} \text{ about } z=0$$

Solⁿ (Result :- for $|z| < 1$, we have
 $(1+z)^{-1} = 1 - z + z^2 - z^3 + z^4 - \dots$ (2))

$$f(z) = \frac{1}{(z+1)(z+2i)}$$

$$= \frac{1}{i(z+1)} + \frac{1}{(1+2i)(z+2i)}$$

$$= -i(z+1)^{-1} + \frac{1}{1+2i}(z+2i)^{-1}$$

$$= -i(1+z)^{-1} + \frac{1}{2i-4} \left(1 + \frac{z}{2i}\right)^{-1} \quad \text{--- (3)}$$

Let $|z| < 1$

$$\text{then } \left| \frac{z}{2i} \right| = \frac{|z|}{2} < \frac{1}{2} < 1$$

So we can apply above result in the brackets of eqn (3)

$$\therefore = -i (1 - z + z^2 - z^3 + z^4 - \dots) + \frac{1}{2i-4} \left(1 - \frac{z}{2i} + \left(\frac{z}{2i}\right)^2 - \left(\frac{z}{2i}\right)^3 + \dots \right)$$

This method of obtaining series form can be used in any type of question to find the series expansion of any function.

Laurent's series

If $f(z)$ is analytic in an annulus R , $(R_1 < |z-a| < R_2)$ with radii R_1 & R_2 & centre a then $\forall z \in R$, the Laurent's series about a is given by

$$f(z) = \underbrace{\sum_{n=0}^{\infty} a_n (z-a)^n}_{\text{Regular part}} + \underbrace{\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}}_{\text{Principal part.}}$$

where $a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(t)}{(t-a)^{n+1}} dt$

$$b_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(t)}{(t-a)^{-n+1}} dt$$

Eg Find Laurent's expansion of $f(z) = \frac{1}{z(z-1)^2}$ (16)
 $\Rightarrow$ in powers of $(z-1)$.

Solⁿ we have $f(z) = \frac{1}{z(z-1)^2}$

$$= \frac{1}{(-1+1+z)(z-1)^2}$$

$$= \frac{1}{[1+(z-1)]((z-1)^2)}$$

~~Result: ③~~ ~~$(1+z)$~~ we want to apply
 result ②, for which we have to
 consider the annulus $0 < |z-1| < 1$.

$$\therefore f(z) = \frac{1}{(z-1)^2} [1+(z-1)]^{-1} \quad \text{--- ①}$$

Since $|z-1| < 1 \therefore$ ① can be written

$$\text{as, } f(z) = \frac{1}{(z-1)^2} [1 - (z-1) + (z-1)^2 - (z-1)^3 + (z-1)^4 - \dots]$$

$$= \underbrace{\frac{1}{(z-1)^2} - \frac{1}{(z-1)}}_{\text{Principal part}} + \underbrace{1 - (z-1)^2 + (z-1)^3 - \dots}_{\text{Regular part}}$$

This is the required Laurent's expⁿ.
 of $f(z)$ in the annulus $0 < |z-1| < 1$.
 Again this method is valid everywhere to
 obtain Laurent's series expansion

eg. $f(z) = \frac{1}{z(z^2-3z+2)}$ Find Laurent's expansion in the annulus $1 < |z| < 2$.

Solⁿ $f(z) = \frac{1}{z(z-1)(z-2)}$

$$= \frac{1}{2z} - \frac{1}{(z-1)} + \frac{1}{2(z-2)}$$

$$= \frac{1}{2z} - \frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} - \frac{1}{4} \left(1 - \frac{z}{2}\right)^{-1}$$

$$= \underbrace{\frac{1}{2z} - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right)}_{\text{Principal part}} - \frac{1}{4} \underbrace{\left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right)}_{\text{Regular part}}$$

$$\left\{ \because |z| > 1 \Rightarrow \left|\frac{1}{z}\right| < 1 \right.$$

$$\text{and } |z| < 2 \Rightarrow \left|\frac{z}{2}\right| < 1.$$

So applying Result (2) which states

$$\text{for } |z| < 1 \quad (1-z)^{-1} = 1 + z + z^2 + z^3 + \dots$$

② Essential singularity If in the Laurent's expⁿ of $f(z)$ about z_0 the principal part contains infinite terms then $z = z_0$ is called essential singularity. [in above example $z=0$ is essential singularity]

(d) Poles of in Laurent's expⁿ about z_0 (17)
 the coefficients of the principal part are 0
 after a particular term b_n . then z_0
 is called pole of order n .

A pole of order 1 is called simple pole.

Eg Find the type of singularity of f^m

$$f(z) = \frac{1 - e^{2z}}{z^3} \text{ at } z = 0.$$

$$\underline{\text{Sol}^n} \quad \frac{1 - e^{2z}}{z^3} = \frac{1}{z^3} \left[1 - \left(1 + 2z + \frac{4z^2}{2!} + \dots \right) \right]$$

$$= - \left(\frac{2}{z^2} + \frac{2}{z} + \frac{4}{3} + \frac{2}{3}z + \dots \right)$$

$$= \underbrace{-\frac{2}{z^2} - \frac{2}{z}}_{\text{principal part}} - \underbrace{\frac{4}{3} - \frac{2z}{3} - \dots}_{\text{regular part}}.$$

after $\underline{b_2}$,
 all coefficients of principal part are 0.
 and series is about the pt. 0.

$\therefore$ 0 is the pole of order 2 ($n=2$)

Residue of a complex function

Defⁿ The coeff b_1 of $\frac{1}{z - z_0}$ in the
 Laurent's expansion of $f(z)$ about the
 singularity z_0 is called the residue of
 $f(z)$ at z_0

In above ~~exp~~ example -2 is the residue of $\odot \frac{1-e^{2z}}{z^3}$ about $z=0$.

OR (Residue (b1), in terms of pole)

$$\textcircled{1} \quad b_1 = \lim_{z \rightarrow z_0} [(z-z_0) f(z)]$$

if z_0 is a simple pole of $f(z)$.

$\textcircled{2}$ If z_0 is a pole of order m then

$$b_1 = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} \{ (z-z_0)^m f(z) \}$$

Eg. Find the residue of $f(z)$ at its pole inside the circle $|z|=2$, where $f(z) = z^2 \cdot e^{1/2}$.

$$\begin{aligned} \underline{\text{Sol}^n} \quad f(z) &= z^2 \left[1 + \frac{1}{z} + \frac{(1/z)^2}{2!} + \frac{(1/z)^3}{3!} + \dots \right] \\ &= z^2 + z + \frac{1}{2} + \frac{1}{3! \cdot 2} + \frac{1}{4! \cdot 2^2} + \dots \quad \textcircled{1} \end{aligned}$$

Since there are infinite no. of -ve powers of z $\therefore z=0$ is essential singularity. $\therefore b_1 = \text{coeff. of } \frac{1}{z} \text{ in Laurent's expansion}$ $\textcircled{1}$

$$\therefore b_1 = \frac{1}{3!} = \frac{1}{6}$$

Ex. Find residue of $f(z) = \frac{1-e^{2z}}{z^3}$ using poles. (18)

solⁿ Since as discussed, $f(z)$ has a pole of order 2 at $z=0$.

$$\therefore \text{Residue } b_1 = \frac{1}{(2-1)!} \lim_{z \rightarrow 0} \frac{d}{dz} (z-0)^2 \cdot f(z)$$

$$= \lim_{z \rightarrow 0} \frac{d}{dz} \cdot z^2 \left(\frac{1-e^{2z}}{z^3} \right)$$

$$= -2.$$

Shortcut method to find singularities.

Consider $\frac{f(z)}{g(z)}$ then

① α is a pole of order m of $\frac{f(z)}{g(z)}$

if α is a zero of order m of $g(z)$ and $f(\alpha) \neq 0$.

② If α is zero of $f(z)$ of order n & α is a zero of $g(z)$ of order m then

$n > m \Rightarrow \alpha$ is removable singularity of $f(z)/g(z)$.

$n < m \Rightarrow \alpha$ is a pole of order $m-n$ of $f(z)/g(z)$.

eg. $f(z) = \frac{\cos z - 1}{z^3}$

$h(z)$ is of the form $\frac{f(z)}{g(z)}$, where
 $f(z) = \cos z - 1$ & $g(z) = z^3$.

$f(0) = 0$		$g(0) = 0$
$f'(0) = 0$		$g'(0) = 0$
$f''(0) \neq 0$		$g''(0) = 0, g'''(0) \neq 0$

$\Rightarrow 0$ is a zero of order 2 of $f(z)$

($\therefore m=2$)

0 is a zero of order 3 of $g(z)$

($\therefore m=3$)

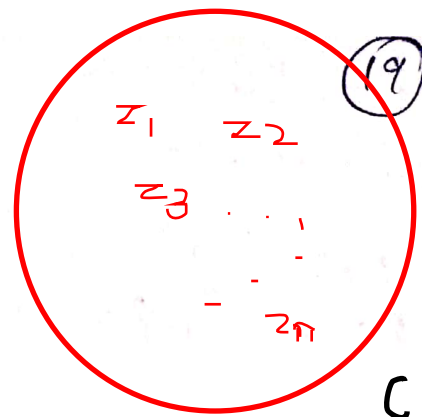
Hence by previous result (2) 0 is a pole of order $3-2=1$ of funcⁿ $h(z)$.

$\hookrightarrow$ we will use above method to find the singularities, as a rough work in the questions based on "Cauchy's residue theorem".

$\hookrightarrow$ Cauchy's residue theorem :- If $f(z)$ is analytic at points in a simple closed curve C except for a finite no. of isolated singularities $z_1, z_2, \dots, z_n$ inside C then

$$\oint_C f(z) \cdot dz = 2\pi i \times (\text{sum of residues of } f(z) \text{ at } z_1, z_2, \dots, z_n).$$

Q. Evaluate $\int_C \frac{dz}{(z-1)(z-2)(z-3)}$
 $C: |z| = 5/2$



Soln let $h(z) = \frac{1}{(z-1)(z-2)(z-3)}$

It is of the form $\frac{f(z)}{g(z)}$
 then by above shortcut method, $z=1, z=2, z=3$ are poles of order 1 of $\frac{f(z)}{g(z)}$

$$\therefore \text{Res. } h(z)_{z=1} = \lim_{z \rightarrow 1} (z-1) \cdot h(z)$$

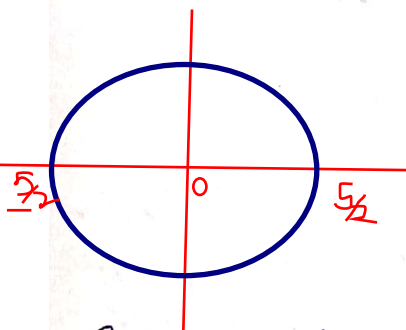
$$= \lim_{z \rightarrow 1} \frac{(z-1) \cdot 1}{(z-1)(z-2)(z-3)} = \frac{1}{2}$$

Similarly $\text{Res. } h(z)_{z=2} = -1$ & $\text{Res. } h(z)_{z=3} = \frac{1}{2}$

$$\therefore \int_C h(z) \cdot dz = 2\pi i \left(\frac{1}{2} - 1 \right) = -\pi i$$

C

we'll not consider residue at $z=3$ $\because z=3$ lies outside the circle $|z| = 5/2$ & Cauchy's residues theor. will not be valid here.



Ex. Evaluate $\int_C \frac{z^2 \cdot dz}{(z+1)^2(z-2)}$ $C: |z-1| = 3$

Soln Again by previous method, $z=-1$ is a pole of order 2 & $z=2$ is simple pole.

$x = -1, x = 2$ lie inside the circle ~~$|z+1|=3$~~
 $|2-1| = 3$.

$$\begin{aligned} \text{Res.}_{z \rightarrow -1} \frac{e^z}{(z+1)^2(z-2)} &= \lim_{z \rightarrow -1} \frac{d}{dz} \frac{(z+1)^2 e^z}{(z+1)^2(z-2)} \\ &= \lim_{z \rightarrow -1} \frac{d}{dz} \frac{e^z}{(z-2)} = -\frac{4}{9e} \end{aligned}$$

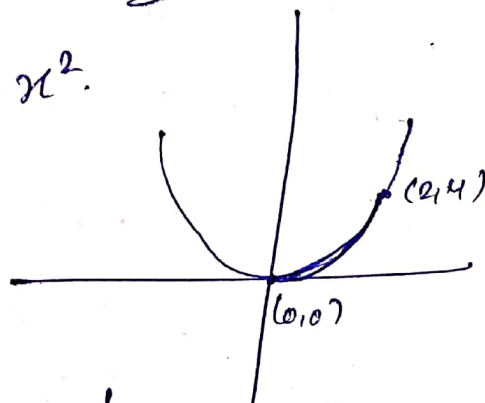
$$\text{Res.}_{z \rightarrow 2} \frac{e^z}{(z+1)^2(z-2)} = \lim_{z \rightarrow 2} (z-2) \cdot \frac{e^z}{(z+1)^2} = \frac{e^2}{9}$$

By Cauchy's residue theorem

$$\oint_C \frac{e^z}{(z+1)^2(z-2)} = 2\pi i \left(\frac{e^2}{9} - \frac{4}{9e} \right) = \frac{2\pi i}{9} \left(e^2 - \frac{4}{e} \right)$$

Ex Evaluate the integral $I = \int_C \text{Re } z^2 dz$
 from $z=0$ to $z=2+4i$
 along the parabola $y=x^2$.

Soln $I = \int_C \text{Re } z^2 dz$



$$= \int_C (x^2 - y^2) (dx + i dy) \quad \left| \begin{array}{l} y = x^2 \\ dy = 2x dx \\ 0 \leq x \leq 2 \\ 0 \leq y \leq 4 \end{array} \right.$$

$$= \int_0^2 (x^2 - x^4) (dx + i 2x dx)$$

$$= \frac{-56}{15} - \frac{40i}{3}$$

Eg Evaluate the integral $\int \bar{z} dz$ along the right half of the circle $|z| = 2$.

Soln. Let $z = re^{i\theta}$.

$$|z| = r = 2.$$

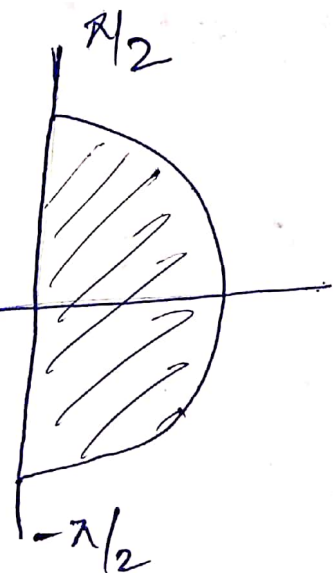
$$\therefore z = 2e^{i\theta}$$

$$\bar{z} = 2e^{-i\theta}$$

$$dz = 2ie^{i\theta} d\theta.$$

$$I = \int_{-\pi/2}^{\pi/2} 2e^{-i\theta} \cdot 2ie^{i\theta} d\theta.$$

$$= 4i \int_{-\pi/2}^{\pi/2} d\theta = 4\pi i.$$



Evaluation of real definite integrals

① Integration around the unit circle

An integral of the type $\int_0^{2\pi} f(\sin\theta, \cos\theta) d\theta$ where the integrand is a rational function of $\sin\theta$ & $\cos\theta$, can be evaluated

by writing $z = e^{i\theta}$.

Eg. Show that $\int_0^{2\pi} \frac{\cos 2\theta}{1 - 2a\cos\theta + a^2} d\theta = \frac{2\pi a^2}{1 - a^2}$, $a^2 < 1$.

Soln. Here $f(\sin\theta, \cos\theta) = \frac{\cos 2\theta}{1 - 2a\cos\theta + a^2}$

let $z = e^{i\theta}$. then $|z| = 1$
 we know, $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

$$\therefore \sin \theta = \frac{z - 1/z}{2i}$$

$$\& \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + 1/z}{2}$$

$$\& \text{So } f(z) \Rightarrow \cos 2\theta = \frac{e^{2i\theta} + e^{-2i\theta}}{2} = \frac{z^2 + 1/z^2}{2}$$

$$\therefore f(z) = \frac{(z^2 + 1/z^2)/2}{1 - 2a\left(\frac{z + 1/z}{2}\right) + a^2} = \frac{z^4 + 1}{2z^2(1 - az - \frac{a}{z} + a^2)}$$

$$= \frac{z^4 + 1}{2z^2 - 2az^3 - 2az + 2z^2a^2}$$

$$= \frac{z^4 + 1}{2z^2(z-a)(1-az)}$$

$$\begin{cases} z = e^{i\theta} \\ \Rightarrow dz = ie^{i\theta} d\theta \\ \Rightarrow dz = iz \cdot d\theta \end{cases}$$

$$\therefore \int_0^{2\pi} f(\sin \theta, \cos \theta) d\theta = \frac{1}{2} \int_C \frac{z^4 + 1}{z^2(z-a)(1-az)} \frac{dz}{iz}$$

$$= \int_C \frac{(z^4 + 1)}{2i(z^2(z-a)(1-az))} dz, \text{ where } C: |z| = 1.$$

Now we apply Cauchy's residue theorem,

Since $f(z)$ has a simple poles at (21)
 $z=a$, $z=1/a$ and a pole of order 2
 at $z=0$.

& $z=0$, $z=a$ lie inside the circle
 $|z|=1$ ($\because$ given $a^2 < 1 \Rightarrow -1 < a < 1$.)

$$\text{Res. } f(z) = \lim_{z \rightarrow a} [(z-a) \cdot f(z)]$$

$$= \frac{a^4 + 1}{2ia^2(1-a^2)}$$

$$\text{Res. } f(z) = \lim_{z \rightarrow 0} \frac{d}{dz} (z^2 \cdot f(z)) = -\frac{(1+a^2)}{2ia^2}$$

$$\therefore \int_C f(z) = 2\pi i \left(\frac{a^4 + 1}{2ia^2(1-a^2)} - \frac{(1+a^2)}{2ia^2} \right)$$

$$= \frac{2\pi a^2}{1-a^2} \text{ Ans. By Cauchy's residue th.}$$

Eg. By integrating around a unit circle

evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta$ Ans $\pi/12$

② Integration around a small semi-circle.

To evaluate the integral of the form
 $\int_{-b}^b f(x) \cdot dx$, we consider $\oint_C f(z)$, where C

is a contour consisting of the semicircle $C_R: |z|=R$ together with the diameter that closes it. R is sufficiently large

Eg. Evaluate $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)}$

Solⁿ Consider $\int_C \frac{z^2 dz}{(z^2+1)(z^2+4)} = \int_C f(z) \cdot dz$

where C is a semi circle C_R of radius R ~~together~~ together with the part of the real axis from $-R$ to R

clearly integrand has simple poles

at $z = \pm i, z = \pm 2i$

of which $z = i$

& $2i$ lie inside

semi circle ($\because R$ is sufficiently large

so points must lie

Applying in upper half plane

only)

Cauchy's residue

thm. to evaluate

$\int_C f(z) \cdot dz$, we get

$$\int_C \frac{z^2 \cdot dz}{(z^2+1)(z^2+4)} = 2\pi i \left[\text{Res. } f(z) \Big|_{z \rightarrow i} + \text{Res. } f(z) \Big|_{z \rightarrow 2i} \right]$$

$$= 2\pi i \left[\frac{1}{6} - \frac{1}{3} \right] = -\frac{\pi}{3} \quad \text{--- (*)}$$

(22)

$$\left[\int_C f(z) \cdot dz = \int_{-R}^R f(x) \cdot dx + \int_{CR} f(z) \cdot dz \quad \text{--- (1)} \right]$$

then make $R \rightarrow \infty$ to find $\int_{-\infty}^{\infty} f(x) \cdot dx$
provided $\int f(z) \cdot dz \rightarrow 0$ as $R \rightarrow \infty$.

we have By property (1).

$$\int_C \frac{z^2 dz}{(z^2+1)(z^2+4)} = \int_{-R}^R f(x) \cdot dx + \int_{CR} \frac{z^2}{(z^2+1)(z^2+4)} dz \quad \text{--- (2)}$$

since $|z|=R \Rightarrow z = Re^{i\theta} \Rightarrow dz = Re^{i\theta} d\theta$

$$\int_{CR} \frac{z^2}{(z^2+1)(z^2+4)} dz = \int_{CR} \frac{R^2 e^{2i\theta} \cdot Re^{i\theta} d\theta}{(R^2 e^{2i\theta} + 1)(R^2 e^{2i\theta} + 4)}$$

$\rightarrow 0$ as $R \rightarrow \infty$ --- (3)

Using (3) in (2) $\int_C \frac{z^2 dz}{(z^2+1)(z^2+4)} = \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)}$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{x^2 \cdot dx}{(x^2+1)(x^2+4)} = \frac{\pi}{3} \quad \text{Ans. (By *)}$$

③ Integration around rectangular contours

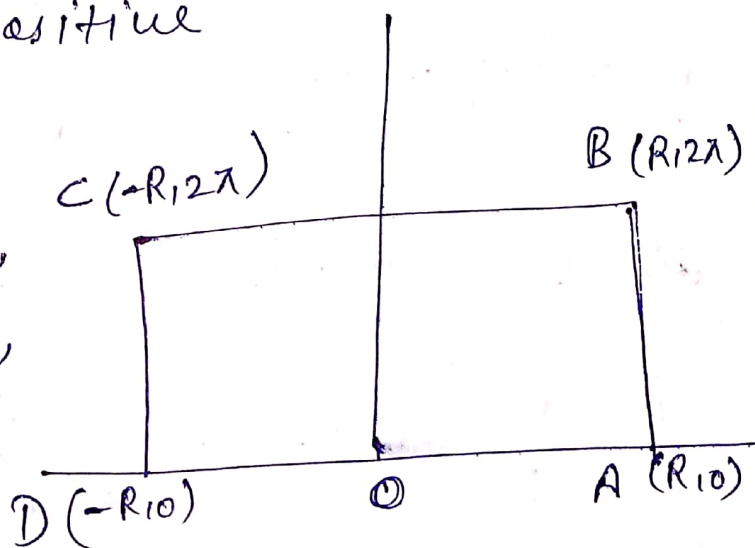
This type of integral is ~~evaluate~~ illustrated with the help of following example.

Eg. Evaluate $\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx$.

Soln Consider $\int_C \frac{e^{az}}{e^z + 1} dz = \int f(z) \cdot dz$.

where C is the rectangle $ABCD$ with vertices at $(R, 0)$, $(R, 2\pi)$, $(-R, 2\pi)$, $(-R, 0)$, R being positive

$f(z)$ has simple poles at $z = (n+1)\pi i$, where $n = 0, \pm 1, \pm 2, \pm 3, \dots$



By Cauchy's residue th.,

$$\int_C f(z) \cdot dz = 2\pi i \left[\text{Res}_{z \rightarrow \pi i} f(z) \right] \quad \left(\because \text{only } z = \pi i \text{ lie inside the rectangle} \right)$$

$$= -2\pi i e^{a\pi i} \quad \text{--- (1)}$$

Also

$$\int_C f(z) \cdot dz = \int_{AB} f(z) \cdot dz + \int_{BC} f(z) \cdot dz + \int_{CD} f(z) \cdot dz + \int_{DA} f(z) \cdot dz$$

--- (2)

Along AB, $0 \leq y \leq 2\pi$, $x = R$ constant. (23)
 So $z = x + iy = R + iy$, $0 \leq y \leq 2\pi$
 along AB.

Along BC, $R \leq x \leq -R$, $y = 2\pi$ constant
 So $z = x + iy = x + 2\pi i$, $R \leq x \leq -R$.
 along BC

Similarly $z = -R + iy$ along CD $2\pi \leq y \leq 0$.
 $z = x$ along DA $-R \leq x \leq R$.

$\therefore$ (2) becomes

$$\begin{aligned} \oint_C f(z) dz &= \int_0^{2\pi} f(R + iy) \cdot i dy + \int_R^{-R} f(x + 2\pi i) \cdot dx \\ &\quad + \int_{2\pi}^0 f(-R + iy) i dy + \int_{-R}^R f(x) dx \\ &= i \int_0^{2\pi} \frac{e^{a(R+iy)}}{e^{R+iy} + 1} dy - \int_R^{-R} \frac{e^{a(x+2\pi i)}}{e^{x+2\pi i} + 1} dx \\ &\quad - i \int_0^{2\pi} \frac{e^{a(-R+iy)}}{e^{-R+iy} + 1} dy + \int_{-R}^R \frac{e^{ax}}{e^x + 1} dx. \end{aligned}$$

(3)

Since $|z_1 + z_2| \geq |z_1| - |z_2|$

$$\therefore |e^{R+iy} + 1| \geq e^R - 1 \quad \text{Also } |e^{a(R+iy)}| = e^{aR}.$$

$$\therefore \left| \frac{e^{a(R+iy)}}{e^{R+iy} + 1} \right| \leq \frac{e^{aR}}{e^R - 1} \rightarrow 0 \text{ as } R \rightarrow \infty. \quad \leftarrow \textcircled{a}$$

$$\text{Similarly } \left| \frac{e^{a(R+iy)}}{e^{-R+iy} + 1} \right| \leq \frac{e^{-aR}}{1 - e^{-R}} \rightarrow 0 \text{ as } R \rightarrow \infty \quad \leftarrow \textcircled{b}$$

Hence $\textcircled{3}$ becomes (when $R \rightarrow \infty$)

$$\int_C f(z) dz = -e^{2a\pi i} \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx + \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx$$
$$= (1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx \quad \leftarrow \textcircled{4} \quad \left(\begin{array}{l} \text{By } \textcircled{a} \\ \text{ \& } \textcircled{b} \end{array} \right)$$

So By $\textcircled{1}$ & $\textcircled{4}$, on equating

$$-2\pi i e^{a\pi i} = (1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx$$

$$\Rightarrow \boxed{\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = \frac{\pi}{\sin a\pi}}$$

Ans